

Mode 10 — Baseline-BFP

C-DP8 (C8×C8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → block exponents

Step 1 — C-DP8 (C8×C8)

$$\text{C-DP8 (C8} \times \text{C8)} = D = \sum_{i=0}^7 a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP16 (8×8)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re} - \sum_{i=0}^7 a_i^{im} b_i^{im}}_{\text{DP16 (8} \times \text{8)}}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im} + \sum_{i=0}^7 a_i^{im} b_i^{re}}_{\text{DP16 (8} \times \text{8)}}$$

Step 3 — DP16 (8×4)

$$\text{Re}(D) = \underbrace{2^4 \sum_{i=0}^7 a_i^{re} b_i^{re,hi} - 2^4 \sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo} - \sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP16 (8} \times \text{4)}}$$

$$\text{Im}(D) = \underbrace{2^4 \sum_{i=0}^7 a_i^{re} b_i^{im,hi} + 2^4 \sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP16 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo} + \sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP16 (8} \times \text{4)}}$$

Step 4 — DP8 (8×4)

$$\text{Re}(D) = \underbrace{2^4 \sum_{i=0}^7 a_i^{re} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} - \underbrace{2^4 \sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}} - \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}}$$

$$\text{Im}(D) = \underbrace{2^4 \sum_{i=0}^7 a_i^{re} b_i^{im,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{2^4 \sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP8 (8} \times \text{4)}}$$

Step 5 — Block exponents (one exponent block per DP8 (8×4) — all copies of one E ; re/im tied, negation scale-blind; parallel matmul: own E_a, E_b)

$$(P_j, E) \text{ per DP8 (8} \times \text{4) above,} \quad E = E_a + E_b \longrightarrow (M^{re}, E), \quad (M^{im}, E)$$

$\delta \equiv 0$: M^{re}, M^{im} are the Step-4 sums unchanged; $\text{Re}(\hat{D}) = M^{re} \cdot 2^E, \text{Im}(\hat{D}) = M^{im} \cdot 2^E$.

Step 6 — Result

$$\begin{aligned} M^{re} &= 2^4 \sum_{i=0}^7 a_i^{re} b_i^{re,hi} - 2^4 \sum_{i=0}^7 a_i^{im} b_i^{im,hi} + \sum_{i=0}^7 a_i^{re} b_i^{re,lo} - \sum_{i=0}^7 a_i^{im} b_i^{im,lo} \\ M^{im} &= 2^4 \sum_{i=0}^7 a_i^{re} b_i^{im,hi} + 2^4 \sum_{i=0}^7 a_i^{im} b_i^{re,hi} + \sum_{i=0}^7 a_i^{re} b_i^{im,lo} + \sum_{i=0}^7 a_i^{im} b_i^{re,lo}, \quad E = E_a + E_b \end{aligned}$$